

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Winter 2022

Date: 09/02/2023

Course : B. Pharmacy
Subject Name : Medicinal Chemistry-I
Max Marks : 75

Sem: IV
Subject Code: BP402T
Duration : 3 Hr.

Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions)

(10 x 2) = 20

- i) Discuss the importance of partition coefficient in relation to biological activity.
- ii) What are neurotransmitters? Classify them with suitable examples.
- iii) Outline the chemical classification of adrenergic drugs with suitable examples.
- iv) Write the MOA of NSAIDs.
- v) Give the biosynthetic pathway of Acetylcholine.
- vi) Draw structure and IUPAC name of Piroxicam.
- vii) What do you mean by tonic clonic seizures?
- viii) Discuss the factors Stereo chemical aspects in relation to drug metabolism.
- ix) Define and distribute adrenergic receptors.
- x) Differentiate generalized seizures & partial seizures

Q. 2. Long Answers (Answer 2 out of 3)

(2 x 10) = 20

- i) Define & classify para sympathomimetic agents with suitable example. Give the SAR, mode of action & chemical features of each class.
- ii) Outline the chemical classification of adrenergic drugs. Discuss their mechanism of action. Comment on the essential structural features required for the optimum activity of such drugs.
- iii) Explain in detailed history and development of medicinal chemistry.

Q. 3. Short Answers (Answer 7 out of 9)

(7 x 5) = 35

- i) Outline the synthesis of diazepam and chlorpromazine hydrochloride
- ii) Classify psychotherapeutic drugs with suitable example. . Discuss the SAR of phenothiazine's as tranquilizing agents.
- iii) Discuss the influence of bioisosterism and partition coefficient on biological activity.
- iv) What are synthetic analgesics? Explain the synthesis and uses of methadone hydrochloride.
- v) Define & classify general anesthetics. Write in brief steps involved in general anesthesia.
- vi) Write a note on biosynthesis and catabolism of catecholamine.
- vii) Discuss on different beta receptor antagonists and write the limitations of non- selective beta blockers.
- viii) What are synthetic analgesics? Explain the synthesis and uses of methadone hydrochloride.
- ix) What are anticholinesterases? Classify & describe the chemistry, MOA & therapeutic uses of the same.